

Network Communication Project

Project Overview

This project explains how network communication takes place when a user opens a secure website. The presentation follows the communication process from the moment a user enters a website address into a web browser through DNS resolution, TCP connection establishment, secure HTTPS communication, data transmission, and delivery of the webpage.

The project also explains how the different layers of the TCP/IP model and OSI model work together with network protocols, IP addresses, MAC addresses, and network devices to support communication across a network.

Scenario

The scenario used in this project is **Opening a Secure Website**.

When a user enters a website address into a browser, several networking processes take place before the webpage is displayed. These include:

1. The browser receives the website address entered by the user.
2. DNS translates the domain name into an IP address.
3. A TCP connection is established between the client and the server.
4. HTTPS provides secure communication between the browser and the web server.
5. Data is divided into packets and transmitted across the network.
6. Routers and other network devices help forward the packets to their destination.
7. The server processes the request and sends the requested webpage back to the user's browser.
8. The browser receives and displays the webpage.

Presentation Topics

The presentation covers the following major areas:

1. Introduction to Network Communication

The introduction explains what network communication means and how devices communicate with one another to exchange data.

2. Application Layer

The Application Layer focuses on the user's interaction with the network. It explains:

- The role of the web browser
- DNS

- Domain names and IP addresses
- HTTP and HTTPS
- How a website request begins

3. Transport Layer

The Transport Layer explains how data is transported between devices.

It covers:

- TCP
- UDP
- TCP port 443 for HTTPS
- TCP three-way handshake
- SYN
- SYN-ACK
- ACK
- Reliable delivery of data

4. Internet Layer

The Internet Layer explains how data is addressed and routed between networks.

It covers:

- IP addresses
- Source and destination IP addresses
- Routing
- Routers
- Movement of packets across networks

5. Network Access and Addressing

This section explains how devices communicate within a local network.

It covers:

- MAC addresses
- Network interfaces
- Local network communication
- Network devices

6. OSI Model Mapping

The project maps the communication process to the OSI model and explains how the different layers relate to the TCP/IP model.

7. Encapsulation and Decapsulation

The presentation explains how data is encapsulated as it moves down the network layers before transmission and how it is decapsulated when it reaches the destination.

8. Network Troubleshooting

The project also examines possible communication problems and how they can be identified and resolved.

One example discussed is:

DNS_PROBE_FINISHED_NXDOMAIN

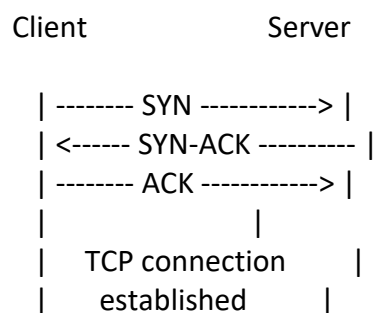
This error can occur when a domain name cannot be resolved to an IP address.

Possible troubleshooting steps include:

- Checking the website address
- Checking the network connection
- Checking DNS settings
- Testing DNS resolution
- Flushing the DNS cache
- Trying another DNS server where appropriate

TCP Three-Way Handshake

Before TCP communication begins, a connection is established using a three-way handshake:



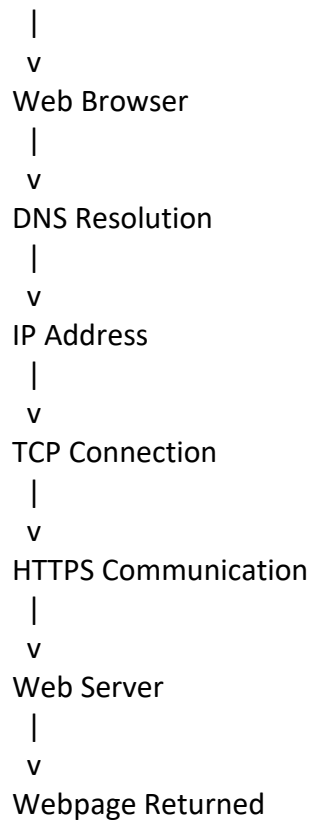
This process allows the client and server to establish a reliable TCP connection before data is exchanged.

Secure Website Communication

For a secure website, HTTPS is used to protect communication between the user's browser and the web server.

The process can be summarized as:

User



Team Contributions

The project was completed collaboratively by five team members. Each member was responsible for a specific part of the network communication process.

Member 1 – Mariam Blessing Ajao

Team Lead - First Speaker

Contribution:

- Introduction to the project
- Network communication overview
- Opening a secure website scenario
- Role of the web browser
- DNS
- HTTP and HTTPS
- Application Layer

Member 2 – Wisdom Binichiuku Odili

Contribution:

- Transport Layer
- TCP

- UDP
- TCP three-way handshake
- TCP and UDP comparison

Member 3 – Beverlyn Ewuradwoa Amanfu

Contribution:

- Internet Layer
- IP addressing
- Routing
- Network devices
- How packets move between networks

Member 4 – Richard Inyang

Contribution:

- OSI model mapping
- Relationship between the OSI and TCP/IP models
- Encapsulation
- Decapsulation

Member 5 – Fokemg-yalie Brisco Elias

Contribution:

- Network troubleshooting
- Common communication problems
- DNS troubleshooting
- Example of DNS_PROBE_FINISHED_NXDOMAIN
- Possible troubleshooting steps

Learning Objectives

At the end of the presentation, the team aims to demonstrate an understanding of:

- The functions of the four layers of the TCP/IP model.
- The TCP/IP layers to the corresponding OSI model layers.
- How to trace data through encapsulation and decapsulation from sender to receiver.
- The roles of IP addresses, MAC addresses, port numbers and protocols during network communication.
- Comparison between TCP and UDP and select the appropriate transport protocol for different applications.
- How to Diagnose a basic TCP/IP communication failure using a logical troubleshooting approach.

- How to communicate technical information clearly and demonstrate accountable teamwork.

Conclusion

The project demonstrates that opening a secure website involves several processes working together. DNS, TCP, HTTPS, IP addressing, routing, network devices, and the different network layers all contribute to successful communication between the user's device and the web server.

Understanding these processes makes it easier to understand how information moves across networks and how common communication problems can be identified and troubleshooted. We used artificial intelligence (AI) search engines to fully understand the concepts of network communication.

Project Evidence

This repository contains the team's project materials and supporting evidence for the Network Communication presentation.

The project demonstrates the collaborative work completed by all five team members and their assigned sections. One of our team members attended the first meeting but was not available for the meetings after that. The other team members and I tried our best to contact the member and also shared the assigned task with the member so that the person could still participate. Unfortunately, we were not able to get a response or have the member join the meetings afterwards.

Despite this, I continued working with the available team members, coordinating the meetings and making sure the project continued moving forward. I also helped put the different team members' contributions together for the final presentation.

Team

Network Communication Project Team

- Mariam Blessing Ajao – Team Lead / First Speaker
- Wisdom Binichiuku Odili – Transport Layer – Second Speaker
- Beverlyn Ewuradwoa Amanfu – Internet Layer – Third Speaker
- Richard Inyang – OSI Mapping and Encapsulation
- Fokemg-yalie Brisco Elias – Troubleshooting– Fifth Speaker